Project Design Phase

Proposed Solution

Date: 12-07-2026

Team ID: SWTID-2026-5429

Project Name: UCab – MERN Stack Cab Booking System

Proposed Solution

S.No.	Parameter	Description
1	**Problem Statement (Problem to be Solved)**	Traditional cab booking methods often involve delays, limited accessibility, and inefficient booking management. Users may face difficulties in finding available cabs, estimating fares, and managing their ride history. There is a need for a secure, user-friendly, and responsive web application that simplifies the cab booking process and provides a seamless user experience.
2	**Idea / Solution Description**	UCab is a full-stack cab booking web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The system enables users to register, log in securely, book cabs by selecting pickup and destination locations, estimate fares, and manage their booking history. The application uses RESTful APIs for communication between the frontend and backend and MongoDB for efficient data storage.
3	**Novelty / Uniqueness**	UCab provides a clean and responsive user interface with secure JWT-based authentication, efficient booking management, fare estimation, and booking history within a single web application. The modular MERN architecture allows easy maintenance and future enhancements such as live GPS tracking, online payments, driver and admin dashboards, and AI-based fare prediction.
4	**Social Impact / Customer Satisfaction**	UCab simplifies transportation by providing a convenient online cab booking platform. It reduces the time required to book rides, improves accessibility, enhances user convenience, and offers a secure booking experience. The responsive interface ensures that users can access the application from desktops, tablets, and mobile devices with ease.

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5	**Business Model (Revenue Model)**	The platform can generate revenue through ride booking commissions, premium membership subscriptions, promotional advertisements, partnerships with local taxi operators, surge pricing during peak hours, and future integration of online payment gateways and promotional offers.
6	**Scalability of the Solution**	The MERN Stack architecture provides high scalability by separating the frontend, backend, and database layers. The application can be easily extended to support thousands of users, cloud deployment (AWS, Azure, Render), driver and admin modules, live GPS tracking, payment gateway integration, push notifications, ride ratings, and advanced analytics without major architectural changes.

Summary

The proposed UCab solution addresses the challenges of traditional cab booking systems by providing a secure, responsive, and efficient web application built using the MERN Stack. Its modular architecture, RESTful APIs, and scalable design make it suitable for future enhancements and real-world deployment while delivering an improved user experience and reliable booking management system.